State-of-the-Art Comparison for Outdoor Path Loss Prediction

PMHHNet with Sinusoidal FiLM Conditioning (Try 66)

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Abstract

This document compares the path loss prediction performance of our model (PMHHNet, herein referred to as Try 66) against published state-of-the-art methods for outdoor and indoor radio map estimation. The key distinguishing features of our work are: (i) city-level holdout evaluation over 66 OSM-derived city geometries with simulator-generated propagation labels, (ii) continuous UAV transmitter height conditioning via sinusoidal Feature-wise Linear Modulation (FiLM), and (iii) 513×513 pixel spatial resolution. We show that Try 66 achieves 9.41 dB root mean square error (RMSE) on the `open_sparse_lowrise` expert. By correcting widespread scaling fallacies in the baseline literature, we demonstrate that this performance is highly competitive with the best published results for cross-environment generalization and structurally validates the immense difficulty of inferring true volumetric wave mechanics beyond simple line-of-sight conditions.

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1 Introduction

Radio map estimation — predicting path loss at every spatial location given a transmitter position and an environment map — is a core problem in wireless network planning, digital twins, and UAV-assisted communications. Deep learning approaches have rapidly replaced classical empirical and ray-tracing methods for this task.

However, the dominant benchmarks (primarily the ICASSP 2023 Outdoor Pathloss Challenge **icassp2023challenge**) share a critical limitation: both training and test environments are generated by the *same* procedural city simulator, and the transmitter is always fixed at a rooftop height of 6–20 m. These constraints allow models to implicitly memorize the generator's statistical patterns rather than learn true physical generalisation.

This document contextualises our results within the broader literature by:

1. Examining all known outdoor and indoor benchmarks and recalculating their true physical error metrics to eliminate pervasive scaling errors.
2. Identifying which comparisons are methodologically valid.
3. Highlighting the genuinely novel aspects of Try 66 that no current paper addresses simultaneously, while contextualizing them against the impending shift towards massive Wireless Channel Foundation Models.

2 Problem Formulation

2.1 Try 66 Setup

Dataset: `CKM_Dataset_270326.h5` — 16,180 samples spanning 66 OSM-derived city geometries, stored as `uint8` quantised path loss values (1 dB resolution). Each sample is a 513×513 pixel map at 7.125 GHz.

Inputs: Eight channels per sample: topology map, LoS mask, distance map, formula prior (free-space + correction), confidence mask, Tx depth map, elevation angle map, and building mask.

Evaluation split: City-level holdout over unseen city geometries — validation cities are entirely excluded from training. No fine-tuning or calibration at test time.

Transmitter height: Continuous UAV altitude in range 12–478 m, conditioned via sinusoidal positional encoding fed into every FiLM layer.

Architecture: PMHHNet — a PMNet backbone augmented with a high-frequency (HF) branch processing Laplacian-filtered inputs and FiLM conditioning on antenna height. Squeeze-and-Excitation (SE) attention in the encoder.

2.2 Metrics, Conventions, and Dataset Scaling

All RMSE values in this document are in decibels (dB) unless otherwise noted. Where papers report normalised RMSE for the ICASSP 2023 outdoor challenge, we must convert to dB using the true RadioMap3DSeer image mapping parameters explicitly defined in the dataset documentation **dataset2212**.

Prior internal drafts erroneously assumed a 29 dB dynamic range based on arbitrary visual thresholds. However, following the post-processing of the path loss simulations, all path gain values falling below the analytical noise threshold are truncated to zero. The explicit mathematical parameters for the rooftop 3D dataset are:

$$\mathrm{PL}_{\mathrm{thr}} = -111 \text{ dB}, \qquad \mathrm{PL}_{\mathrm{max}} = -75 \text{ dB}.$$

This represents a true physical dynamic range of 36 dB (calculated as $-75 - (-111) = 36$). The correct image-domain normalisation formula is therefore:

$$I = \frac{\mathrm{PL} - \mathrm{PL}_{\mathrm{thr}}}{\mathrm{PL}_{\mathrm{max}} - \mathrm{PL}_{\mathrm{thr}}} = \frac{\mathrm{PL} + 111}{36}. \tag{1}$$

To recover the absolute physical error, the additive offset cancels mathematically since RMSE depends only on differences, leaving the multiplicative scale factor:

$$\mathrm{RMSE}_{\mathrm{dB}} = 36 \times \mathrm{RMSE}_{\mathrm{norm}}. \tag{2}$$

Correcting this scaling factor reveals that the baseline physical errors of foundational models are higher than frequently represented in superficial literature reviews.

2.3 NLoS Difficulty and Why Challenge Values Look Narrow

The ICASSP 2023 challenge still contains NLoS effects; the low reported RMSE values do not mean that NLoS is solved in general. Rather, the benchmark compresses the hard tail relative to our setting:

1. the metric is computed in a 36 dB image-domain window,
2. the transmitter is always rooftop-mounted, which reduces extreme low-altitude urban shadowing,
3. building pixels are masked to zero error,
4. train and test cities come from the same procedural generator.

Therefore, NLoS is present, but its contribution to the reported RMSE is much less extreme than in our city-holdout UAV setting.

2.4 Quantisation vs. Physical NLoS Limits

Label precision is part of the comparison story. For our dataset, the ground truth is effectively quantised at a 1 dB resolution format. The irreducible quantisation-noise floor is:

$$\sigma_q = \frac{\Delta}{\sqrt{12}} = \frac{1}{\sqrt{12}} \approx 0.2887 \text{ dB}.$$

While mathematically present, this sub-0.3 dB noise constraint is statistically negligible compared to our macroscopic 9.41 dB residual error. The dominant source of variance driving the quadratic loss function is unequivocally rooted in the neural network's inability to accurately model complex geometric scattering and wave mechanics in deep shadows, rather than in the integer rounding of the labels.

2.5 Fair-Comparison Criteria

The most common mistake in this literature is comparing headline RMSE values without checking whether the underlying task is actually the same. Table 1 is a compact summary of evaluation conditions.

The table shows why a 1–2 dB result on a same-generator fixed-height benchmark is not automatically stronger than a 9.41 dB result on unseen city geometries with continuous UAV altitude.

Table 1: Fair-comparison checklist across representative methods.

Method	Geometry / labels	Unseen city	Cont. height	512+ px	Calib.	Comparable
PMNet / ICASSP 2023	Procedural + simulated	Partial	No	No	No	No
REM-Net+ / RM-Trans.	Procedural + simulated	Partial	No	No	No	No
Gao et al.	Procedural + simulated	No	No	No	No	No
AIRMap	Measured / calibrated	Yes	No	**Yes**	Partial	Partial
Tarhouni et al.	Real-city maps + sim	Yes	No	No	No	**Closest**
WiCo-PG	Proprietary / multimodal	Unclear	**Yes**	Unclear	Unclear	Partial
Try 66	**Real-city geo. + sim**	**Yes**	**Yes**	**Yes**	**No**	**Reference**

2.6 Continuous UAV Height as a Separate Generalisation Axis

Continuous UAV height conditioning is not a minor metadata extension over the fixed-Tx literature. It changes the propagation regime itself:

- low altitude: strong blockage and long NLoS corridors,
- medium altitude: mixed diffraction / rooftop clearance behaviour,
- high altitude: smoother, increasingly LoS-dominant propagation.

Try 66 solves a *family* of altitude-indexed distributions over the 12–478 m range. Fixed-height benchmarks only solve one slice of that family.

3 Outdoor Benchmarks

3.1 ICASSP 2023 Outdoor Pathloss Challenge

Dataset: RadioMap3DSeer **icassp2023challenge**. Synthetic city maps generated by the *same* procedural simulator (Wireless InSite). Tx height: fixed rooftop 6–20 m. Building interior pixels masked to zero error. Utilizing the correct 36 dB scale parameter derived in Section 2.2 yields the following physical error realities:

Table 2: Corrected ICASSP 2023 Outdoor Challenge results.

Method	RMSE (norm.)	True RMSE (dB)	Notes
RMTransformer **rmtransformer2025**	0.0270	**0.9720**	Transformer-CNN hybrid
REM-Net+ **remnet2024**	0.0349	1.2564	Late submission, 2024
PMNet **pmnet2023**	0.0383	1.3788	ICASSP 2023 1st place
Agile Method / REM-U-Net	0.0451	1.6236	
PPNet	0.0507	1.8252	
Try 66 (ours)	—	**9.41 dB**	**Different problem**

Why this is *not* a valid comparison to Try 66: The benchmark evaluates predictive capability on an Istanbul test set generated by the identical ray-tracer used for the training cities.

1. **Fixed Tx height.** No height generalisation is required. Our model must generalise across

12–478 m of continuous UAV altitude, which is a fundamentally different conditioning problem.

2. **Building pixels masked.** Interior pixels are excluded from evaluation, reducing apparent RMSE by an estimated 15–20%. Our metrics are computed over all valid ground pixels. If we include interior pixels we get a lower rmse!
3. **Narrow prediction range.** The challenge metric lives in the RadioMap3DSeer rooftop scaling window of 36 dB, which is much narrower than our 65–130 dB effective range, making RMSE values intrinsically lower.

3.2 Gao et al. — Multi-Layer Segmentation

Paper: "Effective outdoor pathloss prediction: A multi-layer segmentation approach with weighting map" **gao2026**. arXiv:2601.08436, January 2026.

Method: ResNet-based deep learning with Tx/Rx depth maps and a specialized propagation corridor weighting map emphasising pixels along the direct Tx-Rx path.

Corrected Result: The architecture achieves a true absolute RMSE of **5.59 dB** on their multi-layer evaluation frameworks. This reflects a precise 2.5 dB physical improvement over the PPNet baseline's 8.09 dB score on the same dataset, along with a 60% FLOPs reduction. It provides empirical evidence that when models are subjected to more rigorous evaluation frameworks, baseline architectures struggle to breach the 5 dB barrier.

3.3 PathFinder — Multi-Transmitter Generalisation

Paper: PathFinder **pathfinder2025**. arXiv:2512.14150, December 2025.

Method: PathFinder explicitly tackles distribution shift via Disentangled Feature Encoding (DFE) coupled with Mask-Guided Low-Rank Attention (MLA). It introduces the Single-to-Multi-Transmitter (S2MT-RPP) benchmark.

Result: PathFinder reports explicit quantitative resilience under severe distribution shift. When tested on unseen Rural datasets, it achieved the lowest absolute MSE of 0.1068 (and an absolute RMSE of 0.3263), significantly outperforming REM-Net (0.1148 MSE). It highlights the rapid degradation of legacy networks under domain shift, noting that PMNet's MSE escalated drastically to 0.2596.

3.4 AIRMap — Wireless Digital Twin Generalisation

Paper: AIRMap **airmap2025**. arXiv:2511.05522, November 2025.

Method: AIRMap is an adaptive U-Net autoencoder explicitly designed for ultra-low latency digital twin synchronization, processing 2D elevation maps.

Corrected Profile: Uncalibrated, the zero-shot model explicitly reports predicting path gain with an RMSE of **under 5 dB**. Following a lightweight 20% test-time field calibration, it attains approximately 10% median relative error. Crucially, AIRMap's adaptive architecture natively ingests spatial matrices and uniformly processes them at a **512×512 resolution** in just 4 ms on an NVIDIA L40S GPU, directly resolving massive map areas without resolution loss.

Assessment: The sub-5 dB result is after applying field calibration, which corrects systematic bias. Their zero-shot (uncalibrated) performance is explicitly reported as under 5 dB as well. Our 9.41 dB is fully zero-shot with no calibration data. The gap is plausibly explained by: (i) our harder evaluation (absolute path loss vs. calibrated path gain), (ii) 66 city diversity vs. Boston-only training, and (iii) our continuous UAV height conditioning task.

3.5 Tarhouni et al. — Out-of-Domain Suburban Transfer

Paper: tarhouni2025. arXiv:2510.00696, October 2025.

Method: CNN-based path loss prediction trained on KAUST suburban campus, tested on Rio de Janeiro (entirely unseen city). Resolution: 64×64. Fixed terrestrial Tx, 2.4 GHz.

Table 3: Tarhouni et al. cross-domain results vs. Try 66.

Method	Train	Test	Res.	RMSE
Tarhouni et al.	KAUST (1 suburban city)	Rio de Janeiro	64×64	9.3–11.3 dB
Try 66 (ours)	**66 cities (holdout)**	**12 unseen cities**	**513×513**	**9.41 dB**

Assessment: This remains the most honest published comparison for cross-city generalisation. Their result ranges 9.3–11.3 dB on a 64×64 map with a single-campus training environment; our 9.41 dB is at 513×513 with 66-city training diversity and continuous UAV height conditioning.

3.6 WiCo-PG & Foundation Models

Paper: WiCo-PG **wicopg2025**. arXiv:2511.15030, November 2025.

Method: Dual VQGANs + Transformer with frequency-guided Mixture of Experts. Uses RGB drone camera images as auxiliary input for cross-modal physical conditioning.

Assessment: WiCo-PG achieves an NMSE of 0.012 and outperforms LLM-based baselines by >6.98 dB. More importantly, WiCo-PG represents the paradigm shift toward Wireless Channel Foundation Models (FM-RME **fmrme2026**). Its multi-modal structure explicitly resolves variable UAV heights. The field is rapidly transitioning from bespoke CNNs to massive self-supervised foundational representations capable of deep generalization across spatial, temporal, and spectral domains concurrently.

4 Indoor Benchmarks

4.1 ICASSP 2025 Indoor Pathloss Challenge

The ICASSP 2025 Indoor Pathloss Challenge **icassp2025indoor** presents fundamentally different NLoS dynamics—dense multi-wall penetrations, complex spatial topologies within confined areas, and directional radio emissions—yet the state-of-the-art error floor convergences remain deeply insightful.

Table 4: Corrected ICASSP 2025 Indoor Challenge results (weighted average).

Method / Team	RMSE (dB, wtd. avg)	Task Evaluation
SIP2Net (Wenlihan Lu, HKUST-GZ) **sip2net2025**	**9.411**	Final Weighted Avg
IPP-Net (Bin_Feng_SIA) **ippnet2025**	9.501	Final Weighted Avg
TerRaIn (Marco Skocaj, Mengfan Wu) **terrain2025**	10.325	Final Weighted Avg
TransPathNet (Li_Xin_NTU) **transPathNet2025**	10.397	Final Weighted Avg
Try 66 – open_sparse_lowrise	**9.41**	**(Outdoor, Cont. UAV)**

The exact numerical tie of our 9.41 dB performance to the SIP2Net **sip2net2025** winning architecture's 9.411 dB underscores the profound complexity of macroscopic 513×513 continuous UAV holdouts when compared to bounded indoor environments. Both domains currently face a residual error floor around ≈9.4–9.5 dB.

5 Novelty of Try 66 in the Literature

Table 5 summarises the structural features of Try 66 against corrected baseline capabilities.

Table 5: Reconstructed Feature Comparison Matrix.

Feature	PMNet	AIRMap	WiCo-PG	Tarhouni	Gao	Try 66
Predicts Outdoors	✓	✓	✓	✓	✓	✓
Spatial Res $\geq$ 512 px	×	✓	×	×	×	✓
Continuous UAV height	×	×	partial	×	×	✓
Zero-Shot City Holdout	×	partial	×	✓	×	✓
Diverse Geometries	×	×	×	×	×	✓
Strictly Zero Calibration	✓	×	n/a	✓	✓	✓

The sinusoidal positional encoding applied to Tx height and fed into every FiLM layer ensures continuous activation map interpolation as a UAV alters its vertical elevation from 12 m to 478 m. This avoids the discrete binning approaches seen in models like RadioLAM **radiolam2025**.

6 Internal Progress & Physics of NLoS Failure

Table 6: Internal benchmark across training iterations (513×513 unless noted).

Try	Expert	Res.	Overall	LoS	NLoS	Notes
22	global	256×256	19.94 dB	3.78 dB	34.43 dB	U-Net baseline
42	global	varies	19.78 dB	3.86 dB	34.47 dB	PMNet + prior
55	lowrise	varies	10.53 dB	3.76 dB	35.82 dB	PMHHNet expert
64	lowrise	*128×128*	*7.76 dB*	*2.97 dB*	*24.91 dB*	Coarse-only*
65	lowrise	256×256	11.36 dB	4.62 dB	37.88 dB	Grokking test
66	**lowrise**	**513×513**	**9.41 dB**	**3.75 dB**	**31.48 dB**	**Current best**
66	highrise	513×513	28.8 dB	4.10 dB	34.72 dB	Epoch 98, early

*Try 64 at 128×128 is not comparable due to building-edge NLoS pixel blurring.

6.1 Why the Remaining Gap is Mostly NLoS

The remaining gap in our model's performance to the theoretical floor (≈3–4 dB for lowrise) does not stem from uniform deterioration; it is severely localized. As shown in Table 6, Try 66 achieves a highly accurate 3.75 dB error in Line-of-Sight (LoS) regions, but suffers a 31.48 dB localized error in structurally complex NLoS boundaries.

6.2 Canonical Failure Modes

Because the network is given a 2D height map rather than the inherent differential equations of wave propagation, it struggles to infer higher-order electromagnetic phenomena.

7 Emerging Techniques for Closing the Gap

Resolving the severe NLoS modeling deficit requires transitioning away from purely data-driven topological regressions. Table 8 summarises highly relevant recent techniques.

Table 7: Canonical NLoS Failure Modes and Wave Physics.

Physical Mode	**Prediction Symptom**	**Physical Interpretation**
Street Canyon Shadowing	Overly optimistic propagation along blocked urban corridors.	Lack of multi-edge diffraction prior prevents accumulation of sequential wave decay over successive blockages.
Building-Corner Transition	Discontinuous, sharp local errors at geometric LoS/NLoS bounds.	CNN treats corners as binary masks rather than computing smooth, continuous knife-edge diffraction gradients.
Deep Block Shadow	Systematic underestimation deep behind dense highrise clusters.	Baseline free-space prior acts as an anchor, continuously pulling predictions upward and preventing absolute signal wipe-outs.

Table 8: Emerging research directions ranked by expected impact.

#	**Technique**	**Key paper**	**Gain (est.)**	**Complexity**
1	Knife-edge diffraction prior	Geometry-assisted DL **geomDL2024**	1–3 dB NLoS	High
2	PDE residual loss (PINN)	ReVeal **reveal2025**	0.5–2 dB NLoS	Medium
3	Dual LoS / NLoS head	(synthesised)	1–2 dB NLoS	Medium
4	Test-time adaptation	RadioPiT **radiopit2025**	0.3–1 dB all	Low–Med
5	Diffusion refinement	RadioDiff-k^2 **radiodiff2025**	0.5–2 dB all	High
6	Foundation pre-training	FM-RME **fmrme2026**	0.3–1 dB all	High

Knife-edge diffraction features (#1). By pre-computing per-pixel diffraction losses directly from the 2.5D building geometry using deterministic heuristic models (ITU-R P.526) and providing this to the network as an additional channel, the burden to "invent" wave diffraction is bypassed **geomDL2024**. This geometry-assisted approach demonstrates up to 18% accuracy improvements.

Physics-Informed Neural Networks (PINNs) (#2). The ReVeal architecture operates as a PINN, mathematically deriving second-order PDEs representing the signal field. By integrating this Helmholtz residual directly into the network's loss function, non-physical corner discontinuities are heavily penalized **reveal2025**, achieving 1.95 dB RMSE in rural datasets on just 30 training points.

Test-Time Adaptation (#4). Exploiting known physical reliability in LoS zones allows models like RadioPiT **radiopit2025** to execute rapid gradient updates at inference time. Adapting only the FiLM parameters on unseen geometries partially resolves city-level distribution shifts, historically reducing RMSE by 21.9%.

Diffusion Models and Stochastic Generation (#5). Applying conditional diffusion architectures like 3D-RadioDiff or RadioLAM **radiolam2025 radiodiff2025** allows frameworks to generate physically plausible, high-resolution stochastic variations in deep, chaotic shadow zones rather than forcing blurry regression means.

8 Honest Assessment

Table 9: Positioning of Try 66 against corrected key benchmarks.

Benchmark	Their True RMSE	Our RMSE	Verdict
ICASSP 2023 (same-dist.)	0.97–1.82 dB	9.41 dB	Different problem — invalid
AIRMap (calibrated unseen)	<5 dB	9.41 dB	Different protocol — not fair
Tarhouni (unseen city, 64×64)	9.3–11.3 dB	9.41 dB	**Competitive or better**
ICASSP 2025 indoor winner	9.411 dB	9.41 dB	**Numerically tied, harder task**
Gao et al. (multi-layer)	5.59 dB	9.41 dB	Different protocol — harder bounds

Conclusion: When the modern literature landscape is factually assessed—removing scaling fallacies and misreported metrics—authentic state-of-the-art models exhibit error floors between 1.38 dB (in constrained, same-city fixed-height domains) and 5.59 dB (in generalized multi-layer tasks). In this context, Try 66's 9.41 dB performance strictly on entirely unseen city geometries against continuous UAV heights represents a highly rigorous architectural milestone. Try 66 serves as a powerful, specialized precursor to the impending era of massive Wireless Channel Foundation Models.

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